Symbiosis Institute of Technology, Pune

Faculty of Engineering
CSE- Academic Year 2026-27
Java Programming – Lab Batch 2024-28

Lab Assignment No:- 23	
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Batch	A1
Class	CSE
Academic Year & Semester	2026-2027/3RD
Date of Submission	07-10-26
Title of Assignment:	Develop a Java application using JDBC to navigate and retrieve records from a ResultSet.
Theory:	A ResultSet object contains data retrieved from a database after executing a SELECT query. It acts like a cursor that points to one row of the result at a time. Initially, the cursor is positioned before the first row. Various methods such as next(), previous(), first(), last(), absolute(), and relative() allow movement through the records. Data from each column can be retrieved using methods like getString(), getInt(), and getDouble(). Scrollable and updatable ResultSets provide greater flexibility for navigating and modifying retrieved data.
Practice Questions	1. Create a program to display database records one by one using ResultSet.

CODE:

```java
import java.sql.*;

public class StudentRecords
{
    public static void main(String[] args)
    {
        String url = "jdbc:mysql://localhost:3306/school";
        String username = "root";
        String password = "abishek@123";

        try
        {
            Connection con = DriverManager.getConnection(
                url, username, password);
            Statement stmt = con.createStatement();

            String sql = "SELECT * FROM student";
            ResultSet rs = stmt.executeQuery(sql);

            while (rs.next())
            {
                System.out.println("Roll No: " +
                    rs.getInt("roll_no"));
                System.out.println("Name: " +
                    rs.getString("name"));
                System.out.println("Course: " +
                    rs.getString("course"));
                System.out.println("Marks: " +
                    rs.getDouble("marks"));
                System.out.println("--------------------");
            }
            con.close();
        }
        catch (Exception e)
        {
            System.out.println(e);
        }
    }
}
```

OUTPUT:

```
C:\Users\Bina Computer\OneDrive\Documents\Java>javac -cp "lib\
mysql-connector-j-26.7.0.jar" StudentRecords.java
C:\Users\Bina Computer\OneDrive\Documents\Java>java -cp ".;lib\
mysql-connector-j-26.7.0.jar" StudentRecords
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -
Dstderr.encoding=UTF-8
Roll No: 101
Name: Rohan
Course: CSE
Marks: 85.5
--------------------
Roll No: 102
Name: Priya
Course: IT
Marks: 78.0
--------------------
Roll No: 103
Name: Sneha
Course: IT
Marks: 82.0
--------------------
```

QUESTION	2. Create a Java program that retrieves employee records from a database and displays them one by one using ResultSet. Show employee ID, name, department, and salary sequentially.

CODE:

```java
import java.sql.*;

public class EmployeeRecords
{
    public static void main(String[] args)
    {
        String url = "jdbc:mysql://localhost:3306/school";
        String username = "root";
        String password = "abishek@123";

        try
        {
            Connection con = DriverManager.getConnection(
                url, username, password);
            Statement stmt = con.createStatement();

            String sql = "SELECT * FROM employee";
            ResultSet rs = stmt.executeQuery(sql);

            while (rs.next())
            {
                System.out.println("Employee ID: " +
                    rs.getInt("employee_id"));
                System.out.println("Name: " +
                    rs.getString("name"));
                System.out.println("Department: " +
                    rs.getString("department"));
                System.out.println("Salary: " +
                    rs.getDouble("salary"));
                System.out.println("--------------------");
            }
            con.close();
        }
        catch (Exception e)
        {
            System.out.println(e);
        }
    }
}
```

OUTPUT:	C:\Users\Bina Computer\OneDrive\Documents\Java>javac -cp "lib\mysql-connector-j-26.7.0.jar" EmployeeRecords.java C:\Users\Bina Computer\OneDrive\Documents\Java>java -cp ".;lib\mysql-connector-j-26.7.0.jar" EmployeeRecords Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8 Employee ID: 101 Name: Rohan Department: CSE Salary: 55000.0 -------------------- Employee ID: 102 Name: Priya Department: IT Salary: 60000.0 -------------------- Employee ID: 103 Name: Sneha Department: ECE Salary: 62000.0 -------------------- Employee ID: 104 Name: Neha Department: CSE Salary: 58000.0 --------------------

Conclusion	The Java programs successfully retrieved student and employee records from the database using ResultSet. The records were displayed one by one in a sequential manner using the next() method, and the column values were read with getInt(), getString(), and getDouble(). This helped in understanding how ResultSet is used to navigate and retrieve database records.